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United States Patent	12398683
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Kawamura; Taijiro

Injection control apparatus

Abstract

Injection control apparatus includes: injector, drive circuit including battery, booster circuit, and current supply circuit, and controller configured to perform: setting target number of injections and target injection timing per combustion cycle based on operating condition of engine; determining whether first interval from valve opening timing of first injection to valve opening timing of second injection is first threshold value or longer and whether second interval from valve opening timing of second injection to valve opening timing of third injection is second threshold value or longer based on target number of injections and target injection timing, first, second, and third injections being three consecutive injections in one single combustion cycle or two consecutive combustion cycles; and reducing target number of injections when it is determined that first interval is shorter than first threshold value and second interval is shorter than second threshold value.

Inventors:	Kawamura; Taijiro (Tokyo, JP)
Applicant:	Honda Motor Co., Ltd. (Tokyo, JP)
Family ID:	1000008780834
Assignee:	Honda Motor Co., Ltd. (Tokyo, JP)
Appl. No.:	18/823917
Filed:	September 04, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250084804 A1	Mar. 13, 2025

Publication Classification

Int. Cl.: F02D41/40 (20060101); F02D41/30 (20060101)

U.S. Cl.:

CPC **F02D41/402** (20130101); **F02D41/3094** (20130101); F02D2200/021 (20130101)

Field of Classification Search

CPC: F02D (41/402); F02D (41/401); F02D (2041/389); F02D (41/38); F02D (41/40); F02D (41/3094); F02D (41/30); F02D (41/3058); F02D (45/00)

USPC: 123/295; 123/305; 123/456; 123/478; 123/491

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Primary Examiner: Low; Lindsay M

Assistant Examiner: Holbrook; Teuta B

Attorney, Agent or Firm: Duft & Bornsen, PC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2023-146904 filed on Sep. 11, 2023, the content of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention relates to an injection control apparatus configured to control fuel injection of a direct injection internal combustion engine that performs multi-stage injection.

Description of the Related Art

(3) In these years, in order to ensure for all people to have access to affordable, reliable, sustainable, and advanced energy, research and development have been conducted on multi-stage injection of an internal combustion engine for contributing to energy efficiency. As this type of technology, a device that adjusts the number of injections and the injection interval of the multi-stage injection has been conventionally known. For example, in the device described in JP2011-106350 A, the number of injections is set to be capable of ensuring the charging time of a boost capacitor in a high-voltage generation circuit that generates a high voltage to be applied to an electromagnetic solenoid of a fuel injection valve, and in addition, injection at a time later than an injectable time limit is prohibited.

(4) However, in a case where the number of injections is set to be capable of ensuring the charging time of the boost capacitor as in the device described in JP2011-106350 A, the number of injections

of the multi-stage injection is limited, and thereby the effect of the multi-stage injection may be impaired.

SUMMARY OF THE INVENTION

(5) An aspect of the present invention is an injection control apparatus, including: an injector configured to inject fuel into a combustion chamber of an internal combustion engine; a drive circuit including a battery, a booster circuit configured to boost electric power from the battery, and an electric current supply circuit configured to supply drive current to the injector by using the electric power boosted by the booster circuit; and a controller configured to control the drive circuit. The controller includes a processor and a memory coupled to the processor, and is configured to perform: setting a target number of injections and a target injection timing per combustion cycle of the internal combustion engine based on an operating condition of the internal combustion engine; determining whether a first interval from a valve opening timing of a first injection to a valve opening timing of a second injection is equal to or longer than a first threshold value and whether a second interval from the valve opening timing of the second injection to a valve opening timing of a third injection is equal to or longer than a second threshold value based on the target number of injections and the target injection timing, the first injection, the second injection, and the third injection being three consecutive injections in one single combustion cycle or two consecutive combustion cycles; and reducing the target number of injections when it is determined that the first interval is shorter than the first threshold value and the second interval is shorter than the second threshold value.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The objects, features, and advantages of the present invention will become clearer from the following description of embodiments in relation to the attached drawings, in which:
- (2) FIG. 1 is a diagram schematically illustrating a configuration of a substantial part of an engine, to which an injection control apparatus according to an embodiment of the present invention is applied;
- (3) FIG. 2 is a diagram schematically illustrating a configuration of a substantial part of a drive circuit that drives an injector in FIG. 1;
- (4) FIG. 3 is a block diagram illustrating a configuration of a substantial part of the injection control apparatus according to the embodiment of the present invention;
- (5) FIG. 4 is a diagram for describing temperature increase of a piston crown surface in FIG. 1;
- (6) FIG. 5 is a diagram illustrating an example of an injection identification map corresponding to an adhesion reduction mode;
- (7) FIG. 6 is a diagram for describing an injection interval in the adhesion reduction mode;
- (8) FIG. 7 is a diagram for describing an injection interval in a knock suppression mode;
- (9) FIG. 8 is a diagram for describing a boost recovery time of a capacitor in FIG. 1;
- (10) FIG. 9 is a diagram for describing a case where a boosted state of the capacitor is not recovered;
- (11) FIG. 10 is a diagram for describing a case where the boosted state of the capacitor is recovered;
- (12) FIG. 11 is a diagram for describing a modification of first injection, second injection, and third injection; and
- (13) FIG. 12 is a flowchart illustrating an example of processing by the injection control apparatus according to the embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

- (14) Hereinafter, embodiments of the present invention will be described with reference to FIGS. 1

to **12**. An injection control apparatus according to an embodiment of the present invention is applied to a direct injection internal combustion engine that performs multi-stage injection. Hereinafter, in particular, an example in which the injection control apparatus is applied to a direct injection gasoline engine mounted on a vehicle will be described.

(15) FIG. **1** is a diagram schematically illustrating a configuration of a substantial part of an engine **1**, to which an injection control apparatus according to an embodiment of the present invention is applied. The engine **1** is a spark ignition internal combustion engine, and is a four-stroke engine that undergoes four strokes of an intake stroke, a compression stroke, an expansion stroke, and an exhaust stroke during one combustion cycle. The engine **1** includes a plurality of cylinders such as four cylinders, but FIG. **1** illustrates a configuration of a single cylinder. Note that the configurations of the respective cylinders are identical to each other.

(16) As illustrated in FIG. **1**, the engine **1** includes: a cylinder **2a**, which is formed in a cylinder block **2**; a piston **3**, which is slidably disposed inside the cylinder **2a**; and a combustion chamber **5**, which is formed between a crown surface (piston crown surface) **3a** of the piston **3** and a cylinder head **4**. On the piston crown surface **3a**, for example, a recess portion **3b** is formed along a tumble flow in the cylinder. The piston **3** is coupled with a crankshaft **7** through a connecting rod **6**, and the crankshaft **7** rotates, as the piston **3** reciprocates along an inner wall of the cylinder **2a**.

(17) The cylinder head **4** is provided with an intake port **8** and an exhaust port **9**. The combustion chamber **5** communicates with an intake passage **10** through the intake port **8**, and communicates with the exhaust passage **11** through the exhaust port **9**. The intake port **8** is opened and closed by an intake valve **12**, and the exhaust port **9** is opened and closed by an exhaust valve **13**. A throttle valve **14** is provided in the intake passage **10** on an upstream side of the intake valve **12**. The throttle valve **14** includes, for example, a butterfly valve, and an intake air amount into the combustion chamber **5** is adjusted by the throttle valve **14**. The intake valve **12** and the exhaust valve **13** are driven to open and close by a valve mechanism **15**.

(18) An ignition plug **16** and a direct injection injector **17** are attached to the cylinder head **4** so as to face the combustion chamber **5**. The ignition plug **16** is disposed between the intake port **8** and the exhaust port **9**, generates sparks by electric energy, and ignites an air-fuel mixture in the combustion chamber **5**.

(19) The injector **17** is disposed in the vicinity of the intake valve **12**, and is driven by electric energy to inject the fuel. More specifically, the injector **17** is supplied with high-pressure fuel from a fuel tank through a fuel pump. The injector **17** highly atomizes the fuel, and injects the fuel obliquely downward into the combustion chamber **5** at a predetermined timing. Note that the arrangement of the injector **17** is not limited to this, and for example, the injector **17** can be disposed in the vicinity of the ignition plug **16**.

(20) The valve mechanism **15** includes an intake cam shaft **18** and an exhaust cam shaft **19**. The intake cam shaft **18** integrally includes an intake cam **18a**, which corresponds to each cylinder (cylinder **2a**). The exhaust cam shaft **19** integrally includes an exhaust cam **19a**, which corresponds to each cylinder. The intake cam shaft **18** and the exhaust cam shaft **19** are coupled with the crankshaft **7** via a timing belt, not illustrated, and respectively rotate once whenever the crankshaft **7** rotates twice.

(21) The intake valve **12** is opened and closed by the rotation of the intake cam shaft **18** at a predetermined timing in accordance with the profile of the intake cam **18a** via an intake rocker arm, not illustrated. The exhaust valve **13** is opened and closed by the rotation of the exhaust cam shaft **19** at a predetermined timing in accordance with the profile of the exhaust cam **19a** via an exhaust rocker arm, not illustrated.

(22) FIG. **2** is a diagram schematically illustrating a configuration of a substantial part of a drive circuit **20**, which drives the injector **17**. FIG. **2** illustrates an example in which the engine **1** is an in-line four-cylinder engine. In this engine **1**, ignition is conducted whenever the crank angle increases by 180 degrees in the order of a first cylinder #1, a third cylinder #3, a fourth cylinder #4, and a

second cylinder #2. In the following description, it is assumed that an injector **17a** injects the fuel into a combustion chamber **5a** of the first cylinder #1, an injector **17b** injects the fuel into a combustion chamber **5b** of the third cylinder #3, an injector **17c** injects the fuel into a combustion chamber **5c** of the fourth cylinder #4, and an injector **17d** injects the fuel into a combustion chamber **5d** of the second cylinder #2.

(23) As illustrated in FIG. 2, the drive circuit **20** includes: one single booster circuit **22**, which boosts electric power (voltage) from an in-vehicle battery **21**; and an electric current supply circuit **23**, which supplies drive current to the injectors **17a** to **17d** with use of the electric power that has been boosted by the booster circuit **22**. The booster circuit **22** includes one single capacitor **22a**. In this manner, the single booster circuit **22** including the single capacitor **22a** is commonly used by the four injectors **17a** to **17d**, so that the number of component parts can be reduced, the cost can be suppressed, and the configuration can be simplified.

(24) In addition to the capacitor **22a**, the booster circuit **22** includes a booster coil **22b**, an FET **22c** as a switching element, a diode as a rectifier element, a resistor, and the like. One end of the booster coil **22b** is connected with the in-vehicle battery **21**, and the other end of the booster coil **22b** is connected with the drain of the FET **22c**, and is also connected with the capacitor **22a** through a diode. The source of the FET **22c** is connected with a resistor.

(25) When a switching signal is input into the gate of the FET **22c** and the FET **22c** is turned on, a battery voltage is applied to the booster coil **22b**, and electric energy is stored. When a switching signal is input into the gate of the FET **22c** and the FET **22c** is turned off, the electric energy stored in the booster coil **22b** is supplied to and stored in the capacitor **22a**. By repeatedly turning on and off the FET **22c**, the battery voltage is boosted up to a predetermined boosted voltage corresponding to the electrostatic capacity of the capacitor **22a**.

(26) The electric current supply circuit **23** includes FETs **23a**, **23b**, and **23c** as switching elements, diodes as rectifier elements, resistors, and the like. The source of the FET **23a** is connected with an output side of the booster circuit **22**, and the drain of the FET **23a** is connected with one end of a solenoid actuator of each of the injectors **17a** to **17d**. The source of the FET **23b** is directly connected with the in-vehicle battery **21**, and the drain of the FET **23b** is connected with one end of the solenoid actuator of each of the injectors **17a** to **17d** through a diode. The drain of the FET **23c** is connected with the other end of the solenoid actuator of each of the injectors **17a** to **17d**. A diode is interposed between the gate and the drain of the FET **23c**, and a resistor is interposed between the gate and the source of the FET **23c**.

(27) When the injectors **17a** to **17d** are opened from a valve closed state, a switching signal is input into the gates of the FET **23a** and the FET **23c**, which respectively correspond to the injectors **17a** to **17d**, and the FET **23a** and the FET **23c** are turned on. Thus, the capacitor **22a** of the booster circuit **22** is discharged, the boosted voltage is applied to the solenoid actuator of each of the injectors **17a** to **17d**, and each of the injectors **17a** to **17d** is opened from the valve closed state.

(28) After each of the injectors **17a** to **17d** is opened, in a case of maintaining a valve open state, a switching signal is input into the gates of the FET **23a** and the FET **23b**, which respectively correspond to the injectors **17a** to **17d**, the FET **23a** is turned off, and in addition, the FET **23b** is turned on. In this manner, instead of the boosted voltage from the booster circuit **22**, the battery voltage from the in-vehicle battery **21** is applied to the solenoid actuator of each of the injectors **17a** to **17d**, and the valve open state of each of the injectors **17a** to **17d** is maintained.

(29) When the injectors **17a** to **17d** are closed from the valve open state, a switching signal is input into the gates of the FET **23b** and the FET **23c**, which respectively correspond to the injectors **17a** to **17d**, and the FET **23b** and the FET **23c** are turned off. Thus, the supply of the drive current to the solenoid actuator of each of the injectors **17a** to **17d** is stopped, and each of the injectors **17a** to **17d** is closed from the valve open state.

(30) FIG. 3 is a block diagram illustrating a configuration of a substantial part of an injection control apparatus (hereinafter, the apparatus) **100** according to an embodiment of the present

invention. As illustrated in FIG. 3, the apparatus **100** is configured with a controller **30** as its center. The controller **30** is connected with a crank angle sensor **31**, an accelerator opening sensor **32**, a water temperature sensor **33**, an intake air amount sensor **34**, an AF sensor **35**, a fuel pressure sensor **36**, a fuel temperature sensor **37**, and the drive circuit **20**.

(31) The crank angle sensor **31** is provided on the crankshaft **7** (FIG. 1), and is configured to output a pulse signal in accordance with the rotation of the crankshaft **7**. The controller **30** identifies the rotation angle (crank angle) of the crankshaft **7** with reference to the position of the top dead center TDC at the start timing of the intake stroke of the piston **3**, based on a pulse signal from the crank angle sensor **31**, and also calculates the engine speed.

(32) The accelerator opening sensor **32** is provided on an accelerator pedal of a vehicle on which the engine **1** is mounted, and detects an operation amount (accelerator opening) on the accelerator pedal. The target torque of the engine **1** is commanded in accordance with a detection value of the accelerator opening sensor **32**. The water temperature sensor **33** is provided on a passage through which engine cooling water for cooling the engine **1** flows, and detects a temperature of the engine cooling water (cooling water temperature). The intake air amount sensor **34** is a sensor that detects an intake air amount, and includes, for example, an airflow meter disposed on the intake passage **10** (more specifically, on an upstream side of the throttle valve **14**). The AF sensor **35** is provided on the exhaust passage **11**, and detects an air-fuel ratio of the exhaust gas in the exhaust passage **11**.

(33) The fuel pressure sensor **36** is a sensor that detects pressure of high-pressure fuel supplied to the injector **17** via a fuel pump, and is provided, for example, on fuel piping. The fuel temperature sensor **37** is a sensor that detects a temperature of the fuel supplied to the injector **17**, and is provided, for example, on fuel piping.

(34) The controller **30** performs predetermined processing, based on signals from the sensors **31** to **37**, and outputs a switching signal to the FETs **22c** and **23a** to **23c** to control the drive circuit **20**. More specifically, the drive circuit **20** is controlled such that after the injector **17a** injects the fuel into the combustion chamber **5a**, the injector **17b** injects the fuel into the combustion chamber **5b**; after the injector **17b** injects the fuel into the combustion chamber **5b**, the injector **17c** injects the fuel into the combustion chamber **5c**; after the injector **17c** injects the fuel into the combustion chamber **5c**, the injector **17d** injects the fuel into the combustion chamber **5d**; and after the injector **17d** injects the fuel into the combustion chamber **5d**, the injector **17a** injects the fuel into the combustion chamber **5a**.

(35) The controller **30** includes an electronic control unit (ECU), and is configured with a computer including a processor such as a CPU, a memory such as a ROM and a RAM, and other peripheral circuits. The controller **30** includes, as functional configurations, a temperature range determination unit **301**, an injection setting unit **302**, an interval determination unit **303**, and a number of injections correction unit **304**, and functions as the temperature range determination unit **301**, the injection setting unit **302**, the interval determination unit **303**, and the number of injections correction unit **304**.

(36) The temperature range determination unit **301** determines whether the engine **1** is in a low in-cylinder temperature state in which a temperature T_p of the piston crown surface **3a** (FIG. 1) is lower than a predetermined temperature T_{p0} (for example, 100°C .) or in a high in-cylinder temperature state in which the temperature T_p is equal to or higher than the predetermined temperature T_{p0} , based on a workload of the engine **1**. In the case of a gasoline engine, the output (work rate) of the engine **1** has a correlation with an intake air amount G , and the workload (total workload) of the engine **1** has a correlation with an integrated amount ΣG of the intake air amount G . The cylinder **2a** and the piston **3**, which constitute the combustion chamber **5**, have heat capacities corresponding to the respective materials and masses. Hence, a certain amount of heat corresponding to the respective heat capacities, that is, a certain amount of workload is necessary to increase the temperatures of these constituent members.

(37) FIG. 4 is a diagram for describing the temperature increase of the piston crown surface **3a**, and

illustrates an example of a temporal change in soot emission amount to be measured with use of a measuring instrument when the engine **1** is warmed up from a cold state. The temperature T_p of the piston crown surface **3a** indicated in FIG. **4** is an estimated value, and a cooling water temperature T_w is a detection value of the water temperature sensor **33**. In addition, the integrated amount ΣG of the intake air amount G is a calculated value that has been calculated, based on the intake air amount G , which has been detected by the intake air amount sensor **34**.

(38) As indicated in FIG. **4**, in the cold state of the engine **1**, the temperature of the entire engine **1** including the piston crown surface **3a** and the engine cooling water is uniform, and the cold state of the engine **1** is detectable by the water temperature sensor **33**, as the cooling water temperature T_w at the time of starting up the engine **1**. While the engine **1** is being warmed up, the integrated amount ΣG of the intake air amount G (heat amount and workload generated by combustion) increases, and the temperature T_p of the piston crown surface **3a** increases. The temperature T_p of the piston crown surface **3a** increases, then the entire engine **1** including the piston crown surface **3a** is gradually warmed up from the combustion chamber **5** side, and the cooling water temperature T_w increases. The engine **1** is warmed up, then the engine cooling water passes through a radiator, not illustrated, to maintain the cooling water temperature T_w at equal to or lower than a predetermined temperature T_{w0} (for example, 90°C .), and the engine **1** is cooled by the engine cooling water.

(39) As indicated in FIG. **4**, the soot emission amount is substantially constant until time t_0 , and rapidly decreases to become lower than a target emission amount at time t_0 . This point will be described. As illustrated in FIG. **1**, the fuel injected from the injector **17** adheres to the piston crown surface **3a** (recess portion **3b**). In this situation, in a case where the temperature T_p of the piston crown surface **3a** reaches the predetermined temperature T_{p0} (for example, 100°C .), the attached fuel is immediately evaporated, and soot is hardly generated. On the other hand, in a case where the temperature T_p of the piston crown surface **3a** does not reach the predetermined temperature T_{p0} , the adhered fuel is incompletely burned, and soot is easily generated.

(40) By performing a confirmation test of the soot emission amount as illustrated in FIG. **4**, it becomes possible to grasp beforehand an integrated amount (threshold) ΣG_0 of the intake air amount G to be necessitated by the time the temperature T_p of the piston crown surface **3a** reaches the predetermined temperature T_{p0} from the cooling water temperature T_w at the time of starting up the engine **1**. The temperature range determination unit **301** determines that the engine **1** is in the low in-cylinder temperature state, in a case where the integrated amount ΣG of the intake air amount G is smaller than the threshold ΣG_0 , and determines that the engine **1** is in the high in-cylinder temperature state, in a case where the integrated amount ΣG of the intake air amount G is equal to or larger than the threshold ΣG_0 . This eliminates the need for directly detecting the temperature T_p of the piston crown surface **3a**, which reciprocates in the cylinder **2a** while facing the high-temperature combustion chamber **5**, with use of a sensor, and enables determination of whether the temperature T_p of the piston crown surface **3a** has reached the predetermined temperature T_{p0} .

(41) Note that the threshold ΣG_0 as indicated in FIG. **4** varies depending on the cold state of the engine **1**, that is, depending on the cooling water temperature T_w at the time of starting up. That is, the threshold ΣG_0 of the integrated amount ΣG of the intake air amount G to be necessitated by the time the engine **1** reaches the high in-cylinder temperature state increases, as the cooling water temperature T_w at the time of starting up the engine **1** decreases, and decreases, as the cooling water temperature T_w at the time of starting up the engine **1** increases. The characteristic of the threshold ΣG_0 with respect to the cooling water temperature T_w at the time of starting up the engine **1** is stored in a memory beforehand.

(42) The injection setting unit **302** in FIG. **3** sets a target number of injections and a target injection timing per combustion cycle of the engine **1**, based on a predetermined characteristic in accordance with an operating condition of the engine **1**.

(43) Adhesion Reduction Mode

(44) In a case where the temperature range determination unit **301** determines that the engine **1** is in the low in-cylinder temperature state, the injection setting unit **302** sets the target number of injections and the target injection timing so as to suppress the adhesion of the fuel to the piston crown surface **3a**, the inner wall surface of the cylinder **2a**, or the like (adhesion reduction mode). In an adhesion reduction mode, the fuel is injected to generate an air-fuel mixture in a region other than a predetermined injection prohibition region in the vicinity of the intake top dead center TDC at a start timing of the intake stroke and the compression top dead center TDC at an end timing of the compression stroke, that is, in a region where the piston crown surface **3a** is apart from the injector **17** (injectable region). In this situation, by injecting the fuel in multi-stage injection, the fuel injection amount per injection is suppressed. As a result, adhesion of the fuel to the piston crown surface **3a** or the wall surface of the cylinder **2a** can be suppressed, and generation of soot can be suppressed. The injection prohibition region is set to, for example, a partial region or a substantially entire region of an early half of the intake stroke and a partial region or a substantially entire region of a latter half of the compression stroke.

(45) More specifically, the injection inhibition region is set in accordance with the engine rotation speed. As the engine rotation speed is higher, the speed at which the piston crown surface **3a** recedes from the injector **17** in the intake stroke and the speed at which the piston crown surface **3a** approaches the injector **17** in the compression stroke are higher. Therefore, as the engine rotation speed increases, the injection prohibition region in the intake stroke becomes narrower (the end of the injection prohibition region moves to an advance angle side), and the injection prohibition region in the compression stroke becomes wider (the start of the injection prohibition region moves to the advance angle side).

(46) The target number of injections and the target injection timing in the injectable region are determined by an injection identification map stored beforehand in a memory, for example, a characteristic map illustrated in FIG. 5. That is, as illustrated in FIG. 5, they are determined by a predetermined characteristic map in association with a characteristic f of maximum output torque in accordance with an engine speed N_e and a target injection amount Q . In a case where the number of injections is a plurality of times, the injection amounts per injection are equal to each other. Note that the target injection amount Q is calculated as a value at which an actual air-fuel ratio becomes a target air-fuel ratio, and is determined in accordance with the intake air amount G . Therefore, the injection characteristic map of FIG. 5 can be rewritten into a map of the engine speed N_e and the intake air amount G .

(47) Note that in FIG. 5, the crank angle in a section from the start of the intake stroke (intake top dead center TDC) to the end of the compression stroke (compression top dead center TDC) is indicated by an angle of a clockwise circle that starts from the intake top dead center TDC, and the timing of the fuel injection is also indicated by fan-shaped hatching that extends radially from the center of the circle. In the intake stroke, the crank angle is in a range from equal to or larger than 0 degrees to equal to or smaller than 180 degrees, and in the compression stroke, the crank angle is in a range equal to or larger than 180 degrees to equal to or smaller than 360 degrees. A range where the crank angle is equal to or larger than 0 degrees and equal to or smaller than 90 degrees will be referred to as an early half of the intake stroke, and a range where the crank angle is equal to or larger than 90 degrees and equal to or smaller than 180 degrees will be referred to as a latter half of the intake stroke. A range where the crank angle is equal to or larger than 180 degrees and equal to or smaller than 270 degrees will be referred to as an early half of the compression stroke, and a range where the crank angle is equal to or larger than 270 degrees and equal to or smaller than 360 degrees will be referred to as a latter half of the compression stroke.

(48) In order to suppress adhesion of the fuel to the piston crown surface **3a**, it is preferable to increase the number of injections so as to reduce the injection amount per injection. However, a minimum injection amount Q_{min} per injection of the injector **17** is defined in accordance with the

specifications of the injector **17**, and the injector **17** is not capable of injecting an amount smaller than the minimum injection amount Q_{min} (minimum injection time restriction). Therefore, the target number of injections is set to be smaller, as the target injection amount Q is smaller.

(49) In addition, a minimum pause time from closing of the injector **17** to opening of the injector **17** is defined in accordance with the specifications of the injector **17** (minimum pause time restriction). The minimum pause time is set such that a change in the injection amount (flow rate) due to a residual magnetic influence on the solenoid actuator in each of the injectors **17a** to **17d** (FIG. **2**) falls within an allowable range.

(50) On the other hand, in order to increase the number of injections, it is necessary to drive the injector **17** at a high speed, and it is necessary to repeat charging and discharging of the capacitor **22a** in the drive circuit **20** (FIG. **2**) in a short time. In this case, it is necessary to increase the driving speed of the injector **17**, as the engine speed N_e increases. The electric load of the controller **30**, which outputs a switching signal to the FET **22c** of the drive circuit **20**, increases, and the calorific value of the controller **30** increases. As a result, the number of injections is limited by the thermal restriction of the controller **30** (maximum number of injections restriction). Therefore, the target number of injections is set to be smaller, as the engine speed N_e is higher.

(51) Knock Suppression Mode

(52) In a case where the temperature range determination unit **301** determines that the engine **1** is in the high in-cylinder temperature state and a predetermined knock suppression condition is satisfied, the injection setting unit **302** sets the target number of injections and the target injection timing to suppress an occurrence of knocking (knock suppression mode). The knock suppression condition is satisfied when a retard amount of the ignition timing, to be necessary for suppressing an occurrence of knocking at a low rotation and a high load, with respect to the optimum ignition timing is equal to or larger than a predetermined value.

(53) In the knock suppression mode, the retarded ignition timing is returned to the optimum ignition timing side, and the fuel is injected once in the intake stroke (for example, the early half of the intake stroke) and once in the compression stroke (for example, the early half of the compression stroke) (two-stage injection in the intake stroke and the compression stroke). In this case, the injection amount in the compression stroke is the minimum injection amount Q_{min} , and an amount obtained by subtracting the minimum injection amount Q_{min} from the target injection amount Q is injected in the intake stroke. By injecting the fuel in the compression stroke, an end gas temperature of the combustion chamber **5** is decreased by latent heat of vaporization. This enables suppression of an occurrence of knocking, while suppressing the retard amount of the ignition timing. Therefore, the combustion efficiency can be enhanced, as compared with a case where the fuel is injected only in the intake stroke with the retarded ignition timing.

(54) FIG. **6** is a diagram for describing an injection interval Δt in the adhesion reduction mode, and illustrates an injection interval Δt , when the injector **17a** injects the fuel into the combustion chamber **5a** of the first cylinder #1 and subsequently the injector **17b** injects the fuel into the combustion chamber **5b** of the third cylinder #3. The injection interval Δt denotes a period of time between two consecutive injections in a single combustion cycle or two consecutive combustion cycles, and denotes a period of time from the valve opening timing of a previous injection (target injection timing) to the valve opening timing of a next injection.

(55) As illustrated in FIG. **6**, in the adhesion reduction mode in which the multi-stage injection is performed, an injection interval Δt_1 from the final (the fourth time in the illustrated example) injection of the first cylinder #1 (in the illustrated example, injection in the compression stroke) to the first injection of the third cylinder #3 (in the illustrated example, injection in the intake stroke) is shorter than an injection interval Δt_2 from the first injection to the second injection of the third cylinder #3, in some cases. That is, in the adhesion reduction mode, the injection intervals in a single combustion cycle in the identical cylinder are equal to each other. However, the injection interval from the final injection of a previous combustion cycle to the first injection of a next

combustion cycle in two consecutive combustion cycles between the cylinders is shorter than the injection interval in the identical cylinder, in some cases.

(56) FIG. 7 is a diagram for describing the injection interval Δt in the knock suppression mode. As illustrated in FIG. 7, also in the knock suppression mode in which the injection is performed in the intake stroke and the compression stroke, the injection interval Δt_1 from the final injection of the first cylinder #1 (for example, the early half of the compression stroke) to the first injection of the third cylinder #3 (for example, the early half of the intake stroke) is shorter than the injection interval Δt_2 from the first injection to the second injection of the third cylinder #3, in some cases. That is, also in the knock suppression mode, in the two consecutive combustion cycles between the cylinders, the injection interval from the final injection of the previous combustion cycle to the first injection of the next combustion cycle is shorter than the injection interval in the identical cylinder, in some cases.

(57) FIG. 8 is a diagram for describing a boost recovery time of the capacitor 22a (FIG. 2). As illustrated in FIG. 8, after the capacitor 22a is discharged, a predetermined boost recovery time is necessary until the boosted state in which the voltage has been boosted up to the boosted voltage is recovered. More specifically, a predetermined first recovery time Δt_a is needed from the time when the valve of one injector 17 (solenoid actuator) is opened (discharged) in the boosted state to the time when the boosted state is recovered. The first recovery time Δt_a is defined in accordance with the specifications of the injector 17 and the drive circuit 20. After previous discharge is performed, the next discharge is performed in a state in which the boosted state is not recovered, and until the boosted state is recovered, a predetermined second recovery time Δt_b , which is longer than the first recovery time Δt_a , is necessary. The second recovery time Δt_b is the longest in a case where the previous discharge and the next discharge are performed simultaneously. In this case, the second recovery time Δt_b is about twice the first recovery time Δt_a .

(58) When the injection interval Δt_1 is shorter as illustrated in FIGS. 6 and 7, the injection interval Δt_1 is shorter than the first recovery time Δt_a , in particular, at a high rotation and a high load, and the capacitor 22a is insufficiently charged. Thus, the valve opening of the injector 17 is delayed or insufficient, in some cases. In such cases, if the number of injections is reduced so that all the injection intervals Δt become equal to or longer than the first recovery time Δt_a , the effects including a reduction of adhesion of the fuel to the piston crown surface 3a, the inner wall surface of the cylinder 2a, or the like due to multi-stage injection and knock suppression may be impaired. Hence, in the present embodiment, the apparatus 100 is configured as follows so that a limitation to the number of injections of the multi-stage injection can be suppressed.

(59) FIG. 9 is a diagram for describing a case where the boosted state of the capacitor 22a is not recovered, and illustrates the injection intervals Δt of the first injection, second injection, and third injection that are consecutively performed by the injector 17 and the charged state of the capacitor 22a. As illustrated in FIG. 9, when the injection interval Δt_1 from the first injection to the second injection is shorter than the first recovery time Δt_a and the injection interval Δt_2 from the second injection to the third injection is shorter than the second recovery time Δt_b , a voltage drop of the capacitor 22a accumulates. Such accumulation of the voltage drop of the capacitor 22a leads to a significant decrease in the injection amount (a flow rate) of the injector 17, and thus is not allowable. Therefore, it is necessary to reduce the number of injections per combustion cycle to ensure an appropriate injection amount by the injector 17. In this case, for example, it is necessary to reduce the number of injections by prohibiting the final injection in each combustion cycle and to review the injection amount per injection so as to ensure the target injection amount Q.

(60) FIG. 10 is a diagram for describing a case where the boosted state of the capacitor 22a is recovered. As illustrated in FIG. 10, even in a case where the injection interval Δt_1 from the first injection to the second injection is shorter than the first recovery time Δt_a , when the injection interval Δt_2 from the second injection to the third injection is equal to or longer than the second recovery time Δt_b , the voltage drop of the capacitor 22a does not accumulate, and the boosted state

is recovered. In such a temporary voltage drop of the capacitor **22a**, a decrease in the injection amount by the injector **17** is limited. Thus, even though the number of injections is maintained, there is a possibility that a decrease in combustion performance and emission performance due to the decrease in the injection amount falls within an allowable range. For example, in a case where the previous discharge and the next discharge are performed simultaneously, it is possible to determine, in a test beforehand, an operating condition under which the decrease in combustion performance and emission performance due to the decrease in the injection amount falls within an allowable range.

(61) FIG. **11** is a diagram for describing a modification of the first injection, the second injection, and the third injection, and illustrates an injection interval Δt when the two-stage injection is performed. As illustrated in FIG. **11**, the first injection and the second (final) injection in the previous combustion cycle of two consecutive combustion cycles may be set to the first injection and the second injection, and first injection in the next combustion cycle may be set to the third injection. In this case, the previous combustion cycle and the next combustion cycle may be two consecutive combustion cycles in the identical cylinder, or may be two consecutive combustion cycles between different cylinders.

(62) The interval determination unit **303** in FIG. **3** determines whether the injection interval Δt_1 from the first injection to the second injection is equal to or longer than the first recovery time Δt_a , based on the target number of injections and the target injection timing that have been set by the injection setting unit **302**. Then, when it is determined that the injection interval Δt_1 is shorter than the first recovery time Δt_a , it is further determined whether the injection interval Δt_2 from the second injection to the third injection is equal to or longer than the second recovery time Δt_b . The second recovery time Δt_b may be predetermined as, for example, a value about two times the first recovery time Δt_a , or may be calculated in accordance with the injection interval Δt_1 from the first injection to the second injection.

(63) When the interval determination unit **303** determines that the injection interval Δt_2 from the second injection to the third injection is shorter than the second recovery time Δt_b , the number of injections correction unit **304** corrects the number of injections per combustion cycle to be smaller than the target number of injections (in other words, reduces the target number of injections as set by the injection setting unit **302** based on the predetermined characteristic in accordance with the operating condition of the engine **1**). That is, even though the injection interval Δt_1 from the first injection to the second injection is shorter than the first recovery time Δt_a , when the injection interval Δt_2 from the second injection to the third injection is equal to or longer than the second recovery time Δt_b , the target number of injections is maintained. This suppresses a limitation to the number of injections of the multi-stage injection, so that the emission performance and the combustion performance can be improved even with a relatively high rotation and a relatively high load.

(64) The controller **30** calculates the target injection amount Q per combustion cycle in accordance with the intake air amount that has been detected by the intake air amount sensor **34**, while conducting feedback control so that the actual air-fuel ratio that has been detected by the AF sensor **35** becomes the target air-fuel ratio (for example, a theoretical air-fuel ratio). Then, a target injection amount (unit target injection amount) per injection of the injector **17** is calculated in accordance with the target number of injections that has been set by the injection setting unit **302**. More specifically, in the adhesion reduction mode, a value obtained by dividing the target injection amount Q by the target number of injections is calculated as the unit target injection amount, and in the knock suppression mode, an amount obtained by subtracting the minimum injection amount Q_{min} from the minimum injection amount Q_{min} (compression stroke) and the target injection amount Q is calculated as the target injection amount. The controller **30** controls the drive circuit **20** so that the injector **17** injects the fuel of the unit target injection amount at the target injection timing in accordance with the crank angle θ , which has been detected by the crank angle sensor **31**.

That is, by outputting a switching signal to the FETs **22c** and **23a** to **23c** of the drive circuit **20**, the controller **30** controls the drive circuit **20**.

(65) FIG. **12** is a flowchart illustrating an example of processing by the apparatus **100**, and illustrates an example of processing performed by the controller **30** in accordance with a program stored in a memory beforehand. The processing illustrated in this flowchart is started, for example, when the engine **1** starts operating, and is repeated at a predetermined cycle.

(66) As illustrated in FIG. **12**, first, in step **S1**, signals from the sensors **31** to **37** are read. Next, in step **S2**, the target number of injections per combustion cycle and the target injection timing from the injector **17** are set, based on the signals read in step **S1**. Next, in step **S3**, it is determined whether the target number of injections set in step **S2** is a plurality of times, that is, whether the injection is multi-stage injection. In a case where an affirmative determination is made in step **S3**, the processing proceeds to step **S4**, and in a case where a negative determination is made, the processing proceeds to step **S8**.

(67) In step **S4**, it is determined whether the injection interval $\Delta t1$ from the first injection to the second injection is equal to or longer than the first recovery time Δta , based on the target number of injections and the target injection timing set in step **S2**. In a case where a negative determination is made in step **S4**, the processing proceeds to step **S5**, and in a case where an affirmative determination is made, the processing proceeds to step **S8**. In step **S5**, it is determined whether the injection interval $\Delta t2$ from the second injection to the third injection is equal to or longer than the second recovery time Δtb , based on the target number of injections and the target injection timing set in step **S2**. In a case where an affirmative determination is made in step **S5**, the processing proceeds to step **S6**, and in a case where a negative determination is made, the processing proceeds to step **S7**.

(68) In step **S6**, it is determined, based on the signals read in step **S1**, whether the current operating condition of the engine **1** is an operating condition in which a decrease in the combustion performance and the emission performance due to the decrease in the injection amount falls within an allowable range. In a case where an affirmative determination is made in step **S6**, the processing proceeds to step **S8**, and in a case where a negative determination is made, the processing proceeds to step **S7**. In step **S7**, the number of injections is reduced by one, and the processing returns to step **S3**. In step **S8**, the target injection amount and the unit target injection amount are calculated, based on the signals read in step **S1** and the number of injections set and corrected (reduced) in steps **S2** to **S7**, and a switching signal is output to the FETs **22c** and **23a** to **23c** of the drive circuit **20** so that the injector **17** injects the fuel of the unit target injection amount at the target injection timing.

(69) According to the present embodiment, the operations and effects are achievable as follows.

(70) (1) The apparatus **100** includes: the injector **17** configured to inject fuel into the combustion chamber **5** of the engine **1**; the drive circuit **20** including the in-vehicle battery **21**, the booster circuit **22** configured to boost electric power from the in-vehicle battery **21**, and the electric current supply circuit **23** configured to supply drive current to the injector **17**, by using the electric power boosted by the booster circuit **22**; and the controller **30** configured to control the drive circuit **20** (FIGS. **1** to **3**).

(71) The controller **30** includes: the injection setting unit **302** configured to set a target number of injections and a target injection timing per combustion cycle of the engine **1**, based on an operating condition of the engine **1**; the interval determination unit **303** configured to determine whether an injection interval $\Delta t1$ from a valve opening timing of first injection to a valve opening timing of second injection is equal to or longer than a first recovery time Δta and configured to determine whether an injection interval $\Delta t2$ from the valve opening timing of the second injection to a valve opening timing of third injection is equal to or longer than a second recovery time Δtb based on the target number of injections and the target injection timing that have been set by the injection setting unit **302**, in a case where either the number of injections in a single combustion cycle or a total number of injections in two consecutive combustion cycles is at least three times and the three

consecutive injections are defined as the first injection, the second injection, and the third injection; and the number of injections correction unit **304** configured to correct the number of injections per combustion cycle to be smaller than the target number of injections that has been set by the injection setting unit **302** (or configured to reduce the target number of injections as set by the injection setting unit **302**), in a case where the interval determination unit **303** determines that the first interval Δt_1 is shorter than the first recovery time Δt_a and the injection interval Δt_2 is shorter than the second recovery time Δt_b (FIG. 3 and FIG. 12).

(72) That is, even though the injection interval Δt_1 from the first injection to the second injection is shorter than the first recovery time Δt_a , when the injection interval Δt_2 from the second injection to the third injection is equal to or longer than the second recovery time Δt_b , the target number of injections is maintained (FIG. 10). This suppresses a limitation to the number of injections of the multi-stage injection, so that the emission performance and the combustion performance can be improved even with a relatively high rotation and a relatively high load.

(73) (2) The combustion chamber **5** includes the combustion chambers **5a** and **5b**, and the injector **17** includes the injectors **17a** and **17b** configured to respectively inject the fuel into the combustion chambers **5a** and **5b** (FIG. 2). The electric current supply circuit **23** supplies the drive current to each of the injectors **17a** and **17b**, and the controller **30** controls the drive circuit **20** so that after the injector **17a** injects the fuel into the combustion chamber **5a**, the injector **17b** injects the fuel into the combustion chamber **5b**. The first injection denotes a final injection of the combustion cycle by the injector **17a**, the second injection denotes a first injection of the combustion cycle by the injector **17b**, and the third injection denotes a second injection of the combustion cycle by the injector **17b** (FIGS. 6 to 10).

(74) That is, when multi-stage injection is performed by a multi-cylinder engine, in particular, at a high rotation and a high load, the injection interval from the final injection of the previous combustion cycle to the first injection of the next combustion cycle in the two consecutive combustion cycles between the cylinders is decreased, and is shorter than a predetermined boost recovery time, in some cases. Also in such cases, when the injection interval from the first injection to the second injection in the next combustion cycle is sufficiently long, the number of injections is maintained, so that the emission performance and the combustion performance can be improved even with a relatively high rotation and a relatively high load.

(75) (3) The controller **30** further includes the temperature range determination unit **301** configured to determine whether the engine **1** is in a low in-cylinder temperature state in which the temperature T_p of the piston crown surface **3a** is lower than the predetermined temperature T_{p0} or in a high in-cylinder temperature state in which the temperature T_p is equal to or higher than the predetermined temperature T_{p0} (FIG. 3). In a case where the temperature range determination unit **301** determines that the engine **1** is in the low in-cylinder temperature state, the injection setting unit **302** sets the target number of injections and the target injection timing to inject the fuel a plurality of times from the intake stroke to the compression stroke of the combustion cycle (FIG. 6). That is, in the adhesion reduction mode, multi-stage injection is performed to suppress adhesion of the fuel to the piston crown surface **3a**, the inner wall surface of the cylinder **2a**, and the like, so that generation of soot is suppressed and the emission performance is improved.

(76) (4) The controller **30** further includes the temperature range determination unit **301** configured to determine whether the engine **1** is in a low in-cylinder temperature state in which the temperature T_p of the piston crown surface **3a** is lower than the predetermined temperature T_{p0} or in a high in-cylinder temperature state in which the temperature T_p is equal to or higher than the predetermined temperature T_{p0} (FIG. 3). In a case where the temperature range determination unit **301** determines that the engine **1** is in the high in-cylinder temperature state, the injection setting unit **302** sets the target number of injections and the target injection timing to inject the fuel once in each the intake stroke and the compression stroke of the combustion cycle, when a predetermined knock suppression condition is satisfied (FIG. 7). That is, in the knock suppression mode, two-

stage injection in the intake stroke and the compression stroke is performed to reduce an end gas temperature of the combustion chamber 5 with use of the latent heat of vaporization of the fuel that has been injected in the compression stroke, so that an occurrence of knocking is suppressed and the combustion performance is improved, while the retard amount of the ignition timing is being suppressed.

(77) (5) The booster circuit 22 includes one single capacitor 22a (FIG. 2). The plurality of injectors 17a to 17d commonly use the single booster circuit 22 including the single capacitor 22a, so that the number of component parts can be reduced, the cost can be suppressed, and the configuration can be simplified.

(78) In the above embodiment, an example has been described in which the present invention is applied to an in-vehicle gasoline engine. However, the present invention is also applicable to any internal combustion engine other than the in-vehicle internal combustion engine and any internal combustion engine other than the gasoline engine. In addition, although the specific configuration of the injector 17 has been described as an example in FIG. 1 and the like, the configuration of the injection valve that injects the fuel into the combustion chamber is not limited to such an example.

(79) In the above embodiment, the target number of injections and the target injection timing in the adhesion reduction mode are determined in accordance with the characteristic map indicating a relationship between the engine speed and the target injection amount determined in accordance with the intake air amount that has been detected by the intake air amount sensor 34 in FIG. 5 and the like. That is, the engine output torque is detected, based on a signal from the intake air amount sensor 34. However, the injection setting unit that sets the target number of injections and the target injection timing per combustion cycle, based on an operating condition of the internal combustion engine, may be configured to use another physical amount having a correlation with the engine output torque.

(80) In the above embodiment, an example has been described in which the temperature range determination unit 301 determines whether the engine 1 is either in the low in-cylinder temperature state or in the high in-cylinder temperature state by use of the integrated amount of the intake air amount as the workload of the engine 1. However, the temperature range determination unit that determines whether the temperature of the combustion chamber is equal to or higher than the predetermined temperature is not limited to such an example. For example, the integrated amount of the fuel injection amount may be used as the workload of the internal combustion engine. In addition, the temperature of the piston crown surface 3a is estimated as the in-cylinder temperature. However, the temperature of the cylinder wall surface or the rotor surface of the rotary engine may be estimated.

(81) In the above embodiment, an example has been described in which the interval determination unit 303 determines whether the injection interval $\Delta t1$ from the first injection to the second injection is equal to or longer than the first recovery time Δta in step S4 of FIG. 12. In a case where a negative determination is made, the processing proceeds to step S5 to determine whether the injection interval $\Delta t2$ from the second injection to the third injection is equal to or longer than the second recovery time Δtb . However, the interval determination unit that determines whether the first interval is equal to or longer than the first threshold and determines whether the second interval is equal to or longer than the second threshold is not limited to such an example. For example, the determination as to whether the injection interval $\Delta t1$ is equal to or longer than the first recovery time Δta and the determination as to whether the injection interval $\Delta t2$ is equal to or longer than the second recovery time Δtb may be made in parallel with each other. After the determination as to whether the injection interval $\Delta t2$ is equal to or longer than the second recovery time Δtb is made, the determination as to whether the injection interval $\Delta t1$ is equal to or longer than the first recovery time Δta may be made.

(82) The above embodiment can be combined as desired with one or more of the aforesaid modifications. The modifications can also be combined with one another.

(83) According to the present invention, it becomes possible to suppress limitation to the number of injections of the multi-stage injection.

(84) Above, while the present invention has been described with reference to the preferred embodiments thereof, it will be understood, by those skilled in the art, that various changes and modifications may be made thereto without departing from the scope of the appended claims.

Claims

1. An injection control apparatus, comprising: an injector configured to inject fuel into a combustion chamber of an internal combustion engine; a drive circuit including a battery, a booster circuit configured to boost electric power from the battery, and an electric current supply circuit configured to supply drive current to the injector by using the electric power boosted by the booster circuit; and a controller configured to control the drive circuit, wherein the controller includes a processor and a memory coupled to the processor, and is configured to perform: setting a target number of injections and a target injection timing per a combustion cycle of the internal combustion engine based on an operating condition of the internal combustion engine; determining whether a first interval from a valve opening timing of a first injection to a valve opening timing of a second injection is equal to or longer than a first threshold value and whether a second interval from the valve opening timing of the second injection to a valve opening timing of a third injection is equal to or longer than a second threshold value based on the target number of injections and the target injection timing, the first injection, the second injection, and the third injection being three consecutive injections in one single combustion cycle or two consecutive combustion cycles; and reducing the target number of injections when it is determined that the first interval is shorter than the first threshold value and the second interval is shorter than the second threshold value.
2. The injection control apparatus according to claim 1, wherein the combustion chamber includes a first combustion chamber and a second combustion chamber, wherein the injector includes a first injector configured to inject the fuel into the first combustion chamber and a second injector configured to inject the fuel into the second combustion chamber, wherein the electric current supply circuit supplies the drive current to the first injector and the second injector respectively, wherein the controller controls the drive circuit so that the second injector injects the fuel into the second combustion chamber after the first injector injects the fuel into the first combustion chamber, wherein the first injection is a final injection in a combustion cycle of the first injector, wherein the second injection is a first injection in a combustion cycle of the second injector, wherein the third injection is a second injection in the combustion cycle of the second injector.
3. The injection control apparatus according to claim 2, wherein the first injection is an injection in a compression stroke in a combustion cycle of the first injector, wherein the second injection is a first injection in an intake stroke in a combustion cycle of the second injector.
4. The injection control apparatus according to claim 1, wherein the controller is further configured to perform: determining whether a temperature of the combustion chamber is equal to or higher than a predetermined temperature, wherein the controller sets the target number of injections and the target injection timing so that the fuel is injected a plurality of times from an intake stroke to a compression stroke in the combustion cycle when it is determined that the temperature of the combustion chamber is lower than the predetermined temperature.
5. The injection control apparatus according to claim 1, wherein the controller is further configured to perform: determining whether a temperature of the combustion chamber is equal to or higher than a predetermined temperature, wherein the controller sets the target number of injections and the target injection timing so that the fuel is injected once in each an intake stroke and a compression stroke in the combustion cycle when it is determined that the temperature of the combustion chamber is equal to or higher than the predetermined temperature.
6. The injection control apparatus according to claim 1, wherein the booster circuit includes one

single capacitor.

7. The injection control apparatus according to claim 1, wherein the second threshold value is two times the first threshold value.

8. The injection control apparatus according to claim 1, wherein the second threshold value is calculated based on the first interval.
